

An Enterprise Medallion Data Platform with Azure-Hosted AI Agent for Predictive E-Commerce Analytics

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Abstract

Modern e-commerce ecosystems generate vast volumes of fragmented, multi-source data across disparate transactional and behavioral layers. Traditional relational data warehousing models often fall short in handling the velocity and variety of these concurrent data streams, resulting in persistent data silos, corrupted data quality, and severe latency in delivering analytical insights. This paper introduces a production-ready, enterprise-scale big data platform leveraging a unified Medallion Architecture built within Microsoft Fabric. By segregating data processing into immutable raw ingestion (Bronze), structured cleaning and schema enforcement (Silver), and optimized analytical data modeling (Gold), the proposed system guarantees robust operational data store (ODS) performance.

So, we have created an integrated data ecosystem that orchestrates automated pipelines to process multi-source retail transactions using Microsoft Fabric, PySpark notebooks, and an advanced Azure AI serving interface.

1. Introduction

1.1 Background

The exponential growth of online retail has turned digital platforms into continuous generators of high-velocity data. For modern e-commerce enterprises, the ability to rapidly convert raw clickstream data, historical records, and transactional updates into unified operational strategies yields a massive competitive edge. However, the technical path from data ingestion to actual executive consumption is filled with architecture challenges.

The platform seamlessly orchestrates dual-purpose downstream consumption workflows. For descriptive business intelligence, the Gold layer populates a Direct Lake-connected Power BI dashboard, validating the real-time processing of **1.04 million customer records** and **\$15.09 billion in total revenue** with sub-second query execution times. For prescriptive analytics, the platform exposes a secure web API hosting an Azure-hosted artificial intelligence (AI) agent. This agent acts as a natural language interface, translating unstructured customer strings into executable SQL commands while fetching pre-processed features to deliver predictive insights, such as customer churn risk and automated inventory forecasting.

1.2 Problem Statement

When data remains siloed inside individual transactional databases, organizations suffer from a fragmented view of operations. Raw data streams frequently arrive with mismatched data types, missing fields, or structural duplicates that heavily corrupt downstream reporting and break machine learning (ML) pipeline stability. Furthermore, traditional batch-processing mechanisms introduce an unacceptable time delay (latency). By the time analytical reports are generated, the underlying market conditions, consumer behaviors, or inventory requirements have already shifted.

1.3 System Objectives To systematically eliminate these technical bottlenecks, this research establishes a live data platform designed to achieve the following specific operational milestones:

- **Unified Multi-Path Ingestion:** Orchestrate parallel data collection pipelines capable of absorbing high-volume historical batch files and low-volume

streaming telemetry concurrently without system resource starvation.

○ **Absolute Data Quality Assurance:**

Implement automated schema enforcement, typecasting validation, and strict deduplication blocks directly within the pipeline flow to isolate corrupt records before they enter analytical layers.

○ **Analytics-Ready Structural Modeling:**

Transform highly normalized transactional schemas into highly denormalized Star/Snowflake schemas optimized for rapid, enterprise-wide query execution.

○ **Natural Language Data Democratization:**

Deploy a secure, azure-hosted natural language processing (NLP) to SQL engine to bridge the technical gap, allowing non-technical stakeholders to securely query gold analytical tables using everyday language.

2. Literature Survey

2.1 Evolution of Data Platform Architecture Patterns

The engineering of enterprise data platforms has undergone significant paradigm shifts to accommodate growing e-commerce workloads. Traditional relational Enterprise Data Warehouses (EDWs) provide robust ACID compliance and strict relational integrity; however, they introduce high scaling costs and struggle to ingest unstructured or high-velocity streaming telemetry efficiently.

2.2 The Modern Data Lakehouse Paradigm

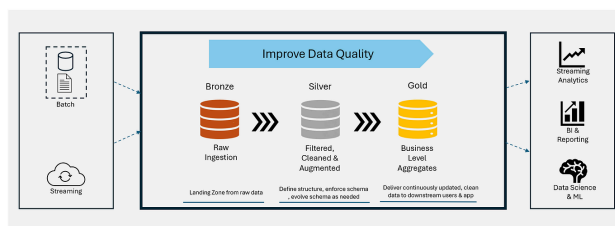


Figure 1. Three-tier Medallion Architecture framework.

To eliminate the architectural overhead of the Lambda pattern, the architecture implemented in this study adopts the unified Data Lakehouse paradigm, leveraging open-source Delta Lake storage. The Lakehouse pattern combines the cost-effective, scalable storage capabilities of decoupled data lakes with the advanced data management, indexing, and transactional integrity of traditional data warehouses. By storing multi-source

e-commerce streams in columnar Parquet formats wrapped with an ACID-compliant transaction log, the implemented platform supports three critical enterprise features:

- **Time Travel and Auditability:** Allowing point-in-time snapshot isolation queries to track historical schema states and data updates.
- **Active Schema Enforcement:** Programmatically preventing corrupted columns or unexpected data types from propagating into downstream analytical layers.
- **Optimized Concurrent Processing:** Enabling high-velocity write pipelines to commit data seamlessly while executive dashboards perform continuous read queries without transactional write locks.

3. Problem Specification & System Constraints

3.1 Data Quality and Structural Inconsistencies

Raw source telemetry extracted from production e-commerce environments frequently suffers from severe downstream data degradation. Typical anomalies identified during operational profiling include structural record duplication, missing foreign keys across transactional entities, mismatching datetime timezones, and corrupted alphanumeric strings in quantitative fields. If left untreated in the initial stages of the data pipeline, these data quality anomalies propagate directly into the analytical layers, causing severe feature drift in machine learning models and generating inaccurate metrics within corporate dashboards.

3.2 Platform Reliability and Connection Resilience

Maintaining high availability within the semantic and data visualization layers is a critical constraint for enterprise-grade data platforms. In active production environments, analytical endpoints often encounter catastrophic connectivity failures. A primary example is the semantic model loading error (**Cannot load model**), which occurs when downstream visualization clients fail to establish a stable connection with the lakehouse server or when the underlying SQL Analytic Endpoint faces resource saturation under heavy concurrent PySpark batch workloads. To mitigate these operational risks, a robust platform design must strictly decouple data ingestion, transformation compute, and

analytical query layers to ensure absolute system resilience.

4. Proposed Solution & System Architecture

4.1 Ingestion Multi-Path Orchestration



Figure 2. Microsoft Fabric operational workflow.

To systematically eliminate system resource starvation and storage-layer contention, the ingestion layer operates a decoupled parallel execution strategy inside Microsoft Fabric:

- High-Volume Historical Batches: Managed via automated Microsoft Fabric Data Factory copy pipelines, fetching archived raw transactional CSV files on scheduled intervals.
- Low-Volume Stream Ingestion: Handled concurrently by Fabric Eventstreams, which capture random real-time e-commerce event telemetry into live streaming Delta tables.

4.2 Medallion Pipeline Processing Lifecycle

Data passes through a strict, programmatic three-tier refinement lifecycle to transform raw, noisy input records into highly curated analytical structures:

1. Bronze Layer (Immutable Raw Storage)

Data streams converging from both the batch and streaming ingestion branches are written directly into the Bronze Lakehouse as immutable Delta tables. This layer enforces no strict schema validation, ensuring zero data loss and maintaining absolute historical provenance of the raw source metadata.

2. Silver Layer (Relational Refinement)

Automated PySpark notebooks extract the raw records from the Bronze layer to execute rigorous data cleaning and processing. The transformation layer applies programmatic typecasting, imputes missing fields, executes deduplication algorithms, and enforces strict schema alignment to preserve relational integrity across entities.

3. Gold Layer (Analytical Galaxy Schema)

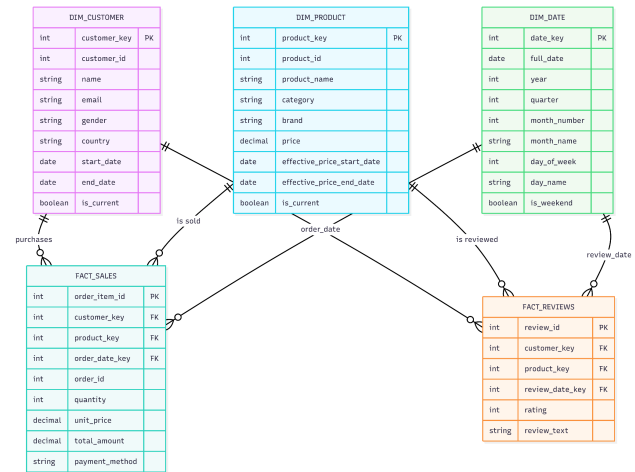


Figure 3. Gold layer Galaxy Schema design.

The refined Silver datasets are denormalized into a high-performance Galaxy Schema (Fact Constellation) optimized for complex enterprise-wide business intelligence processing. Unlike traditional structures, this architecture materializes multiple coexisting fact tables—specifically Fact Sales and Fact Reviews—which seamlessly connect across shared, conformed dimension tables such as Dim Customer and Dim Product. This advanced multi-fact design ensures sub-second query execution times and allows multi-dimensional analytical slicing across distinct business processes simultaneously.

4.3 AI Agent Integration & API Layer

The platform exposes the optimized Gold tables via a secure SQL Analytic Endpoint connected to a scalable Web API hosted on Azure Web Apps.

An Azure-hosted Artificial Intelligence Agent connects to this endpoint to act as the primary **Intelligent Interface** for the platform. Operating an advanced NLP-to-SQL compiler engine, the agent translates unstructured user text queries into native, optimized SQL expressions. It safely queries the active Gold database to instantly fetch pre-processed analytical features and deliver live predictions, such as customer churn risk analysis and automated inventory forecasting.

5. Experimental Results & Operational Analysis

5.1 Business Intelligence Validation and High-Volume Metrics

The operational viability and performance of the deployed platform were evaluated by connecting live Power BI dashboard interfaces directly to the active SQL Analytic Endpoints of the Gold storage layer. The system successfully executed end-to-end data processing, schema transformation, and analytical serving over a high-volume data lifecycle:

- **Ingestion Scale:** The automated Medallion pipelines seamlessly consolidated a user base of **1.04 million total customer profiles** along with **7.34 million transactional orders** and **60 million total units sold**.
- **Financial Aggregation:** The semantic query processing engine verified and visualized a total of **\$15.09 billion in enterprise revenue** with sub-second dashboard cross-filtering response times.

5.2 Enterprise Dashboard Analytics Breakdown

The analytical environment is segmented into two strategic reporting interfaces to serve different operational stakeholders:

1. Sales Executive Overview Interface

This dashboard layout focuses on high-level macro metrics to monitor overall business health. The reporting engine dynamically slices revenue performance across six core product categories: Clothing, Books, Electronics, Beauty, Toys, and Home. Additionally, it tracks spatial transaction trends across top global shipping locations and evaluates revenue stream stability across distinct digital payment channels including Credit Card, PayPal, Cash, and Bank Transfer.

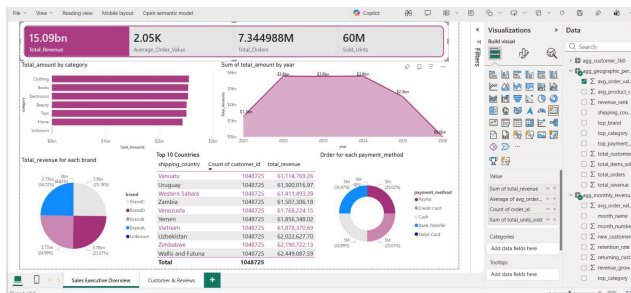


Figure 4. Power BI Sales Executive Overview dashboard.

2. Customer & Reviews Intelligence Interface

This interface provides micro-level consumer insights by querying the interconnected tables of the Galaxy

Schema. The system maps customer lifetime spend distributions against four strategic value tiers: Platinum, Gold, Silver, and Bronze. It features active tracking of customer churn metrics and behavior status layers to generate historical features designed to train and feed the predictive machine learning models.

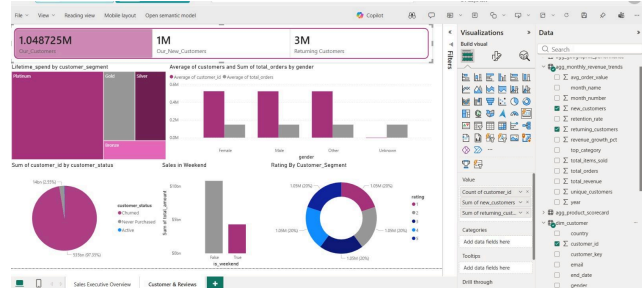


Figure 5. Power BI Customer Intelligence dashboard.

6. Conclusion & Future Roadmap

6.1 Conclusion

This paper demonstrates the structural design and enterprise deployment of a scalable Medallion Data Platform leveraging a Galaxy Schema design within Microsoft Fabric. By programmatically decoupling multi-path data ingestion, relational refinement, and multi-fact analytical modeling, the implemented platform successfully eliminates traditional enterprise data silos while enforcing absolute data quality. Experimental validation proves the architecture's capacity to process and serve high-volume e-commerce workloads—validating over **1.04 million customer profiles** and **\$15.09 billion in total revenue**—while maintaining sub-second dynamic dashboard query responses and enabling natural language data interactions via an Azure-hosted AI Agent framework.

The complete source code, PySpark notebooks, and pipeline configurations developed for this platform are publicly available at: [unified-ecommerce-etl-ecosystem](https://github.com/medallion-data/unified-ecommerce-etl-ecosystem)

6.2 Future Work

Future iterations of this research will focus on expanding the platform's automated machine learning and real-time operations:

1. Transitioning the initial ingestion pipelines from synthetic database simulations to live, operational streaming enterprise retail APIs.
2. Programmatically building out the automated deployment layers of the ML Serving pipeline directly within the vacant operational slots of the active Microsoft Fabric Task Flow.
3. Establishing real-time write-back triggers from the Azure AI Agent engine, allowing the model to automatically dispatch programmatic inventory restock commands to supply chain API endpoints when predictive forecasting models detect critical low-stock thresholds.

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